

Probing the limits of the Universal Models for Atoms: energetic and structural analysis of polyoxometalates

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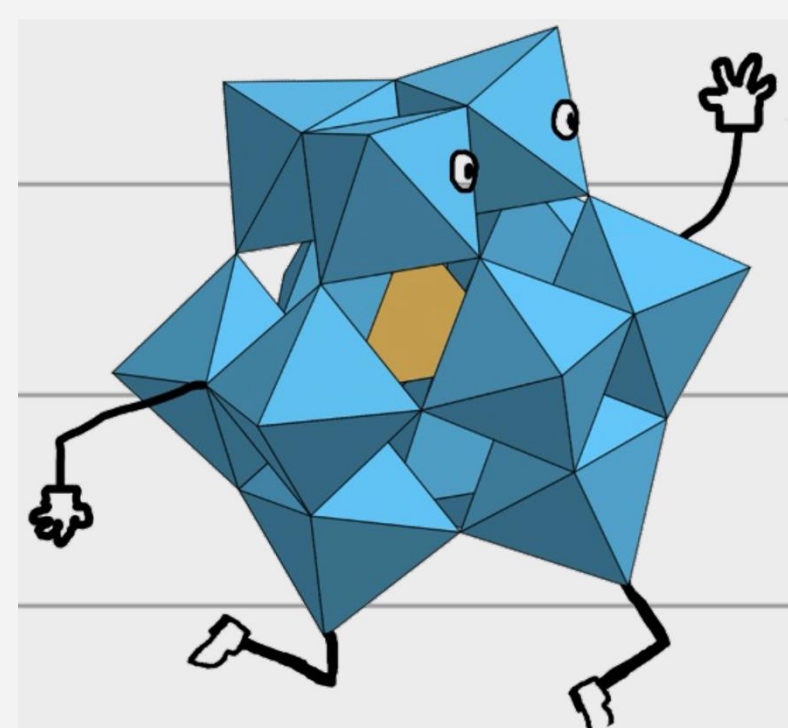
A. Introduction

We developed **UMA-ASE**, a new interface that enables efficient calculations and simulations with the **Universal Model for Atoms (UMA)**. **Polyoxometalates (POMs)** provide a challenging and informative benchmark for this method because their **high charge**, **structural diversity**, and **strong electrostatics** make them difficult systems to model and an excellent test of MLIP transferability.

B. Objective

To evaluate the performance and limitations of the Universal Model for Atoms potential, in polyoxometalates by analyzing:

- **Absolute energies**
- **Relative energetics**
- **Structural deviations (RMSD)**
- **Family-wise energetic ranking with respect to DFT reference calculations.**



C. Methodology

Dataset:

- Canonical **Keggin** isomers (controlled benchmark)
- Extended dataset (~700+ systems)
- Variation in:
 - **Metal composition (W, Mo, Ta)**
 - **Lacunary structures (O40, O39, O34, O31)**
 - **Charge states and protonation**

Calculations:

- **DFT** (AMS, Gaussian16. Gas phase and implicit solvent)
- **UMA via ASE** (gas phase)

Metrics:

- **Absolute energy error**
- **Relative energy error**
- **RMSD (structure)**
- **Spearman correlation (ranking)**

E. Analysis

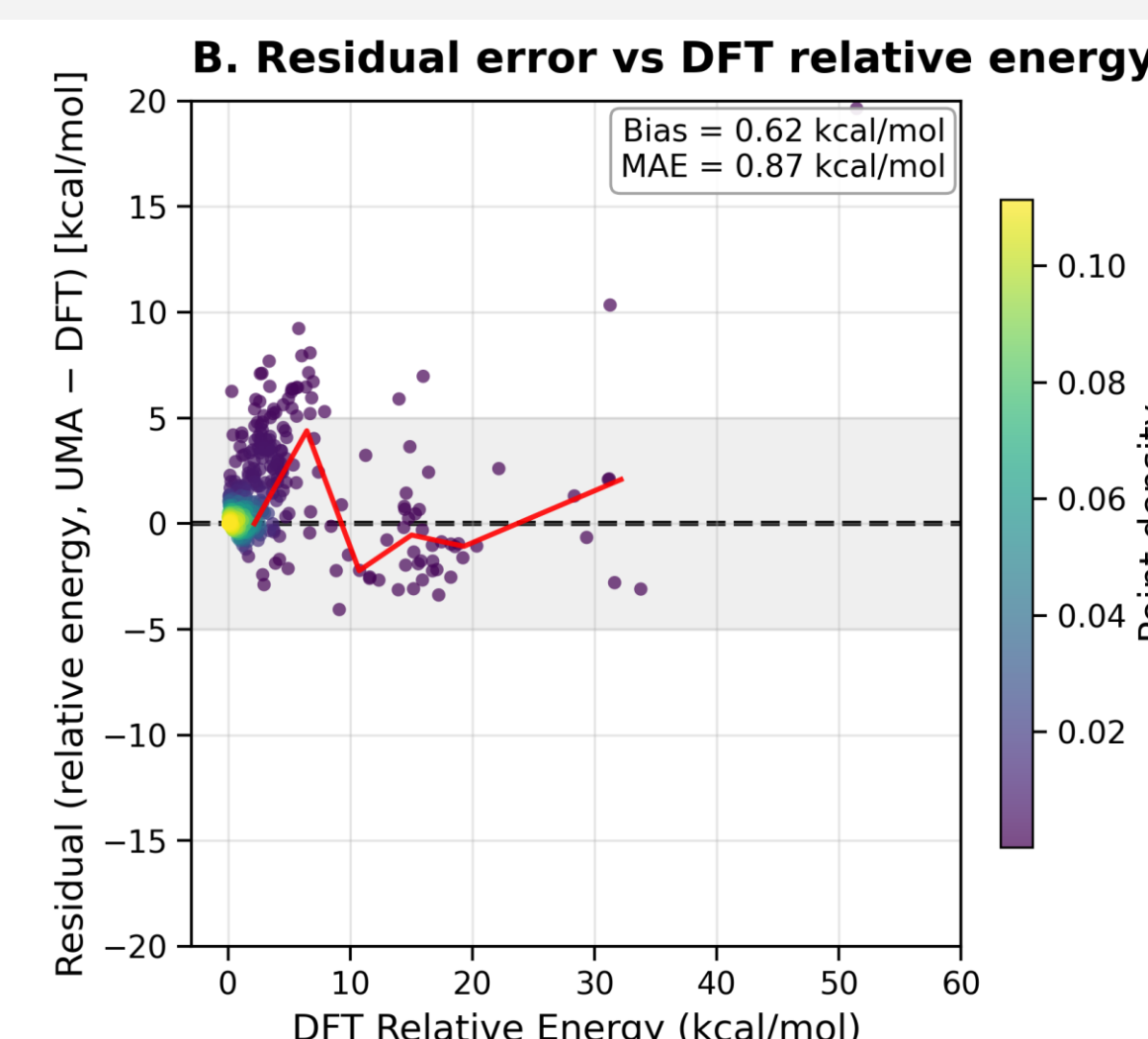
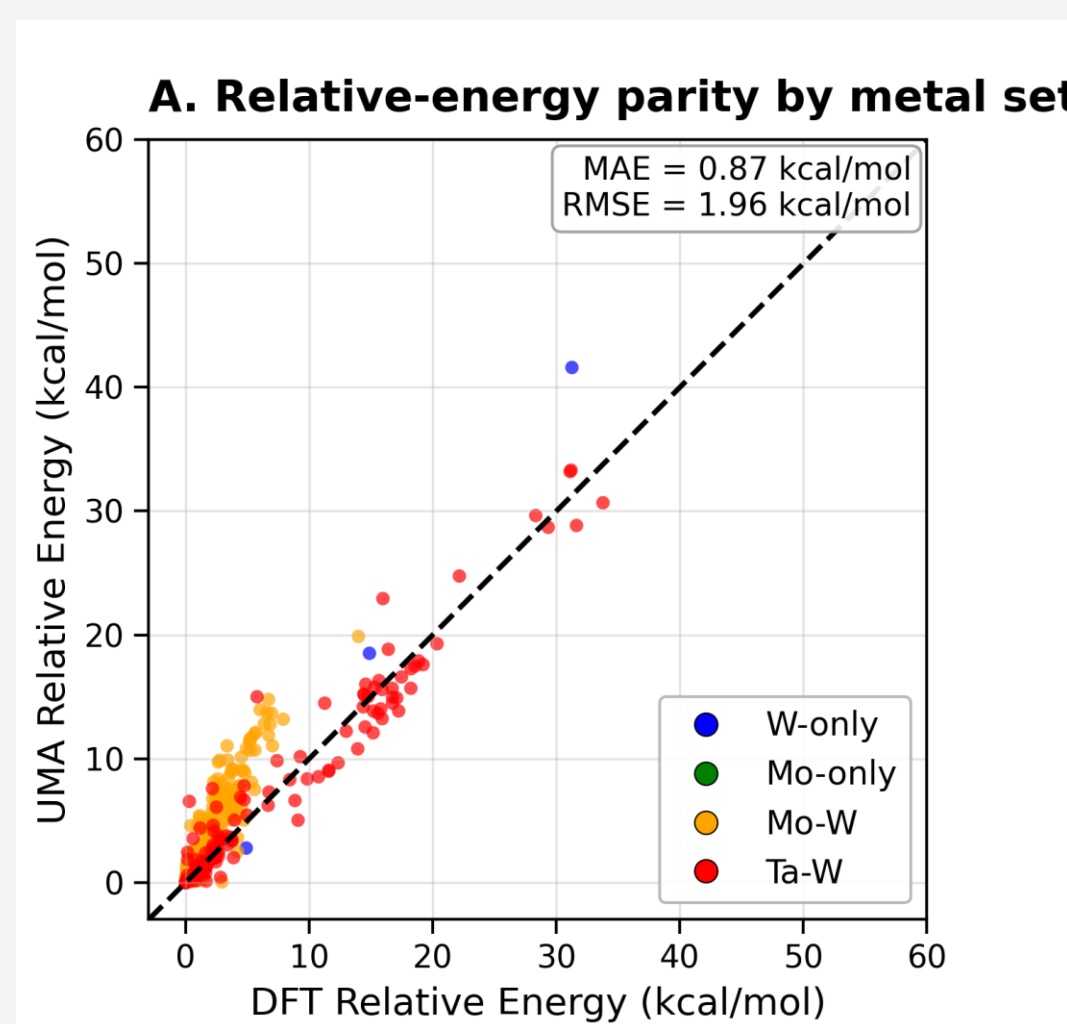
UMA reproduces gas-phase **geometries** and **relative-energies**. **Absolute energies** show large systematic deviations, mainly driven by the poor representation of highly **charged species** in the training set and the lack of **solvent stabilization**. As a result, error cancellation within families enables reliable **energetic ranking**, while absolute energetics remain outside the method's current range of validity.

Key insight: **good structural agreement does not necessarily translate into accurate absolute energetics.**

D. Results

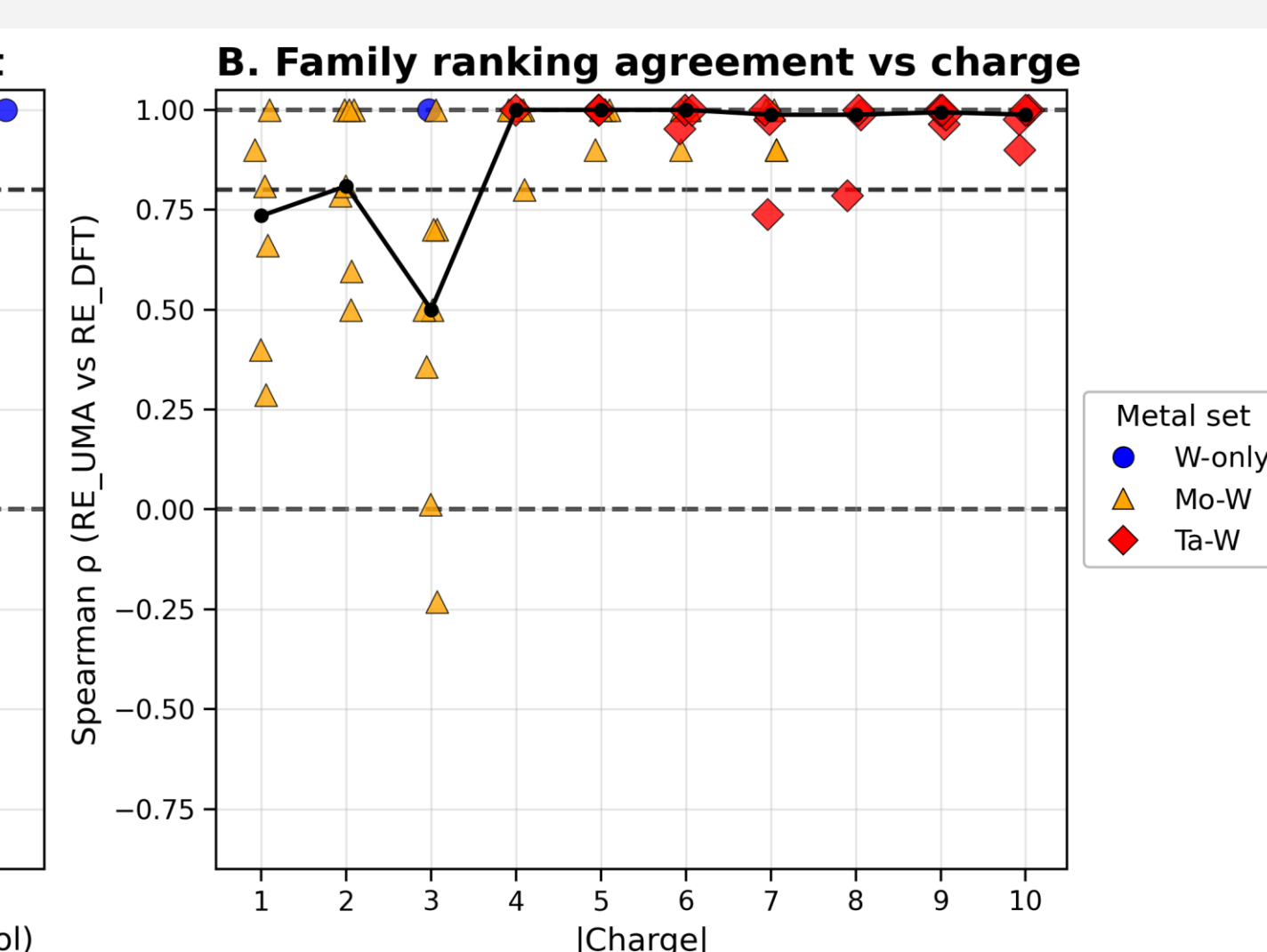
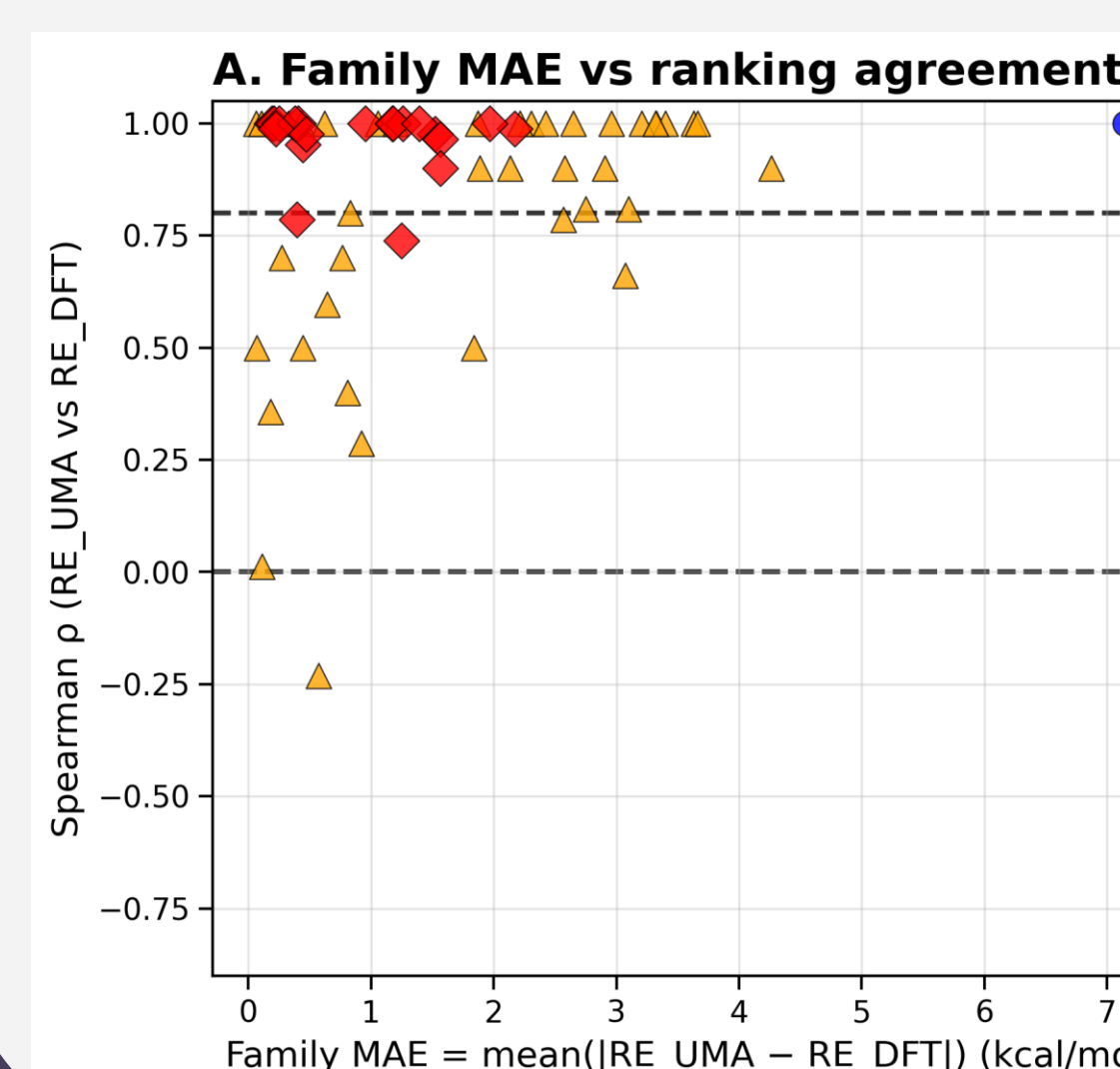
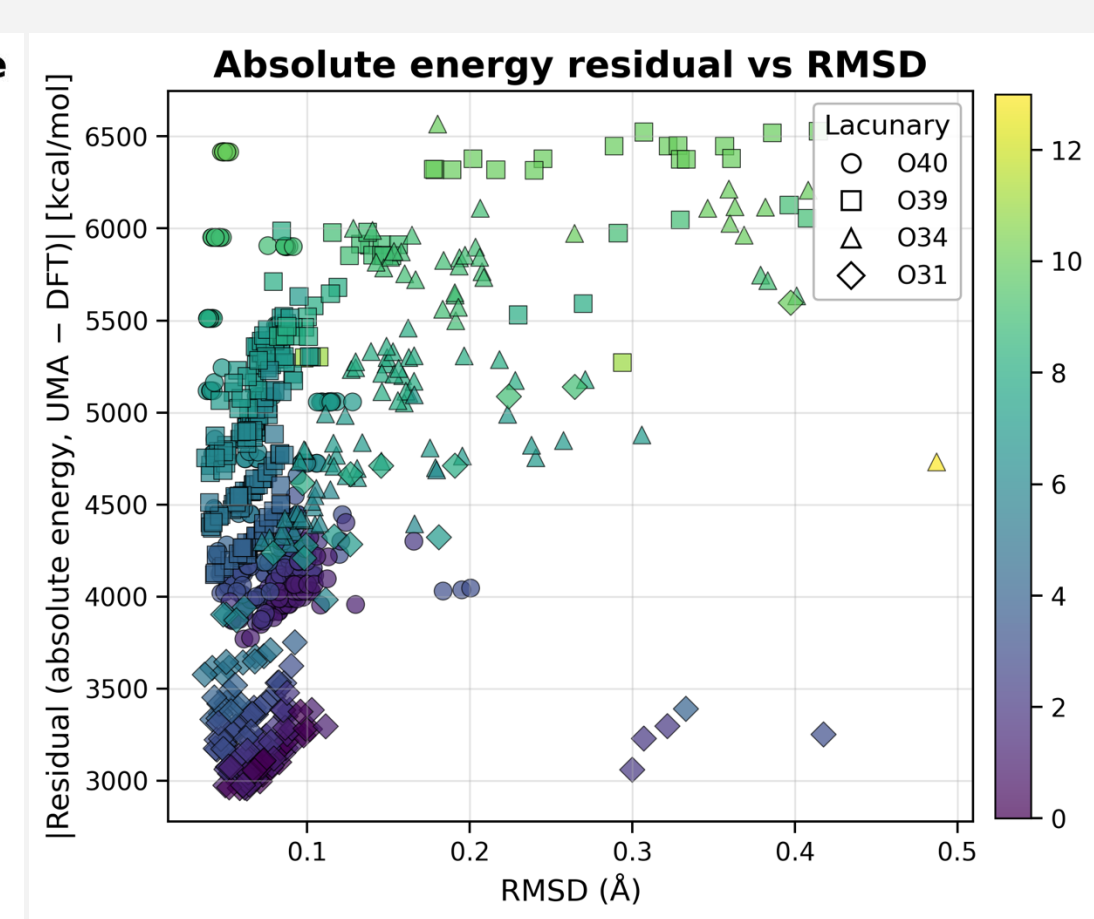
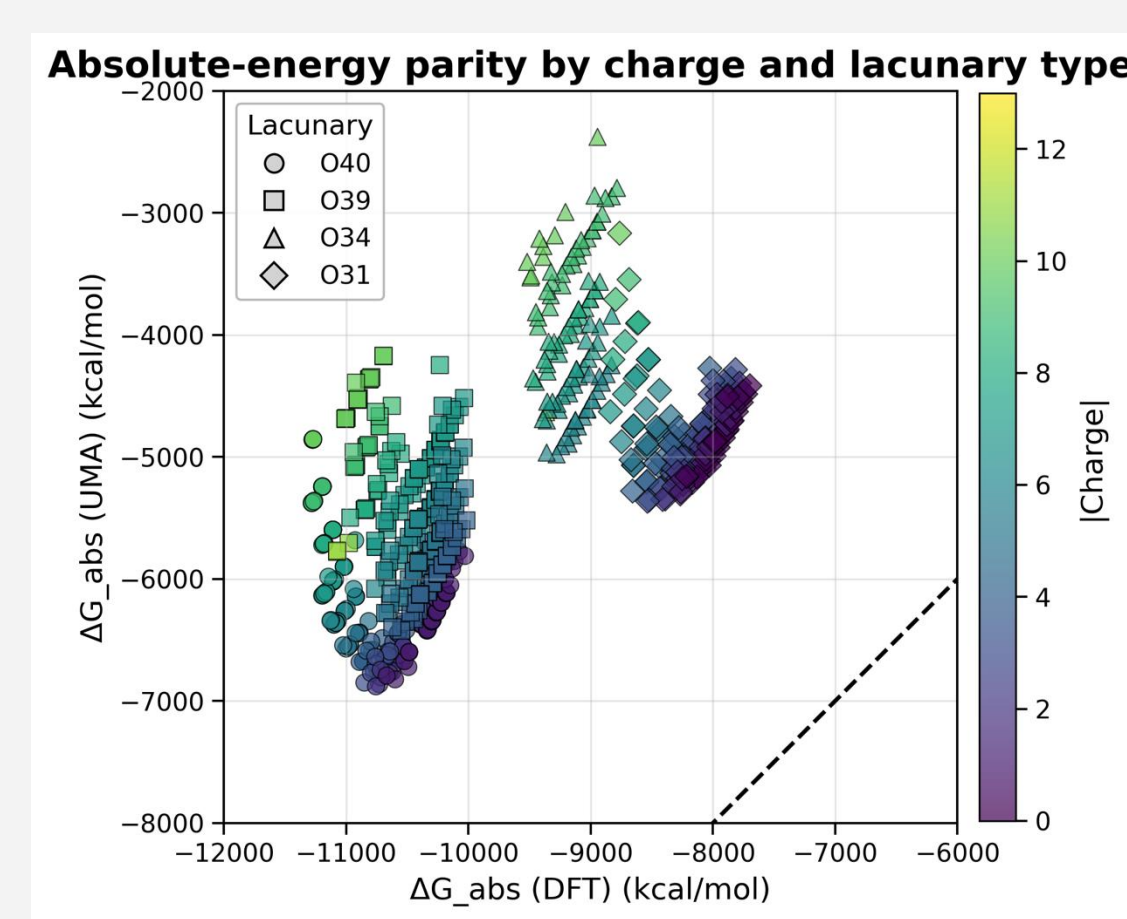
	Total Energy (kcal/mol)		Relative energy (kcal/mol)			RMSD (Å)			TIME		
	Error G16-UMA	% Error G16-UMA	AMS	G16	UMA	AMS-UMA	G16-UMA	AMS-G16	AMS (hrs/min)	G16 (hrs/min)	UMA (sec)
ALPHA	0.578	0.0139	0.0	0.0	0.0	0.08	0.05	0.034	0/59	26/14	7.9
BETA	0.582	0.0140	5.04	5.49	2.88	0.05	0.05	0.009	1/02	28/12	6.5
GAMMA	0.577	0.0139	15.14	17.85	18.65	0.07	0.07	0.008	0/56	28/34	6.4
DELTA	0.572	0.0138	32.63	38.07	41.76	0.08	0.08	0.009	0/59	28/24	6.4
EPSILON	0.565	0.0136	55.56	63.02	71.11	0.12	0.11	0.026	1/03	25/49	6.3

Under gas-phase conditions, **UMA reproduces absolute energies**, while **preserving the relative-energy ordering** across the Keggin isomer series. **Structural agreement is likewise excellent**. UMA delivers these results at a **fraction of the computational cost**, as energies and interatomic forces are obtained directly from the machine-learned potential.



Relative energies are more **accurate** with systematic **errors** largely **cancel** within **families** and also the **energetic trends** are **preserved**.

UMA exhibits large systematic deviations in absolute energies and a strong dependence on system **charge**. Also we can see a clear separation by **lacunary** type and a consistent **internal organization**.



Energetic ordering is well preserved and the limitations are related to **failures** occur when **errors** are **unevenly** distributed and exist a weak dependence on **charge**.

F. Conclusions

- UMA fails to reproduce **absolute energies** in **highly charged** systems.
- **Errors** are **systematic** and dominated by the **lack** of stabilizing **solvation effects**.
- Relative energies are **highly reliable**.
- Energetic trends and **ranking** are preserved across **families**.
- UMA is suitable for: **screening and relative stability analysis**.
- UMA is not reliable for: **Absolute energetics in highly charged clusters**.